IoT, open-data, bought data monitoring

What this tool produces · *Continuous, real-time data streams tracking air quality, waste levels, and traffic flows, translated into actionable public dashboards and automated alerts for municipal services.*

G4 · Proof of Sustainability

Dimensions

Environmental

Spatial

Economic

aktan

With simplexity

Preparing the workshop · *IoT, open-data, bought data monitoring*

G4 · Proof of Sustainability

Why this tool matters

Giving the city a nervous system; transforming silent concrete and asphalt into a living, communicative entity that constantly informs the community about the health of its shared environment.

Evidence we produce

Continuous, real-time data streams tracking air quality, waste levels, and traffic flows, translated into actionable public dashboards and automated alerts for municipal services.

Duration & Material

Duration · Continuous deployment; 3 to 6 months for initial sensor network setup

Material · IoT sensors, edge computing hardware, open data platforms, secure communication networks

Reference case

Chicago’s Array of Things project operates as an urban observatory utilizing edge computing to process environmental and infrastructure data directly at the source. By transmitting only critical summaries instead of raw sensor data, the city monitors its environment efficiently without compromising network capacity. Similarly, Amsterdam uses IoT sensors to optimize waste collection, demonstrating concrete operational savings of up to 40%.

Suggested agenda · Brief (10 min) › Method walk-through (15 min) › Animation canvas (60–90 min) › Harvest (20 min)

Three data-source families · trade-offs

Duration
Continuous deployment; 3 to 6 months for initial sensor…

IoT sensors

live · costly · privacy-sensitive

Open data

free · slow · uneven

Bought data

rich · expensive · contracts

Use case × data source

Use case	IoT	Open	Bought
Mobility			
Air quality			
Public space			
Energy			
Safety			

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.